1. Fundamentals of Data Communication

Data communication is the exchange of data between two nodes via some form of transmission medium.

- **Five Components:** Message, Sender, Receiver, Transmission Medium, and Protocol (the set of rules).
- **Transmission Modes:**
  - **Simplex:** One-way communication (e.g., Television broadcasting).
  - **Half-Duplex:** Two-way, but not at the same time (e.g., Walkie-talkie).
  - **Full-Duplex:** Simultaneous two-way communication (e.g., Mobile phone call).

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2. Transmission Media

The physical path through which data travels is classified into two types:

A. Guided Media (Wired)

- **Twisted Pair (UTP/STP):** Common in LANs (RJ45 connectors). Cheap but susceptible to interference.
- **Coaxial Cable:** Used for cable TV and older networks. Better shielding than twisted pair.
- **Fiber Optic:** Uses light pulses to transmit data. Offers the highest bandwidth, longest distance, and is immune to electromagnetic interference.

B. Unguided Media (Wireless)

- **Radio Waves:** Used for Wi-Fi and Bluetooth; omnidirectional.
- **Microwaves:** Used for line-of-sight communication (Satellite and Terrestrial towers).
- **Infrared:** Short-range, line-of-sight (Remote controls).

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3. Network Topology

Topology refers to the physical or logical layout of a network.

Topology	Description	Advantage	Disadvantage
Star	All nodes connect to a central Hub/Switch.	Easy to install; one failure doesn't affect others.	Central hub failure kills the network.
Bus	All nodes share a single main cable (backbone).	Cheap and uses less cabling.	If the backbone breaks, the whole network fails.

Topology	Description	Advantage	Disadvantage
Ring	Nodes form a closed loop; data travels in one direction.	No data collisions.	Adding/removing a node disrupts the network.
Mesh	Every node is connected to every other node.	Highly redundant and reliable.	Very expensive and complex to cable.

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4. The OSI Reference Model

The ISO developed the **Open Systems Interconnection (OSI)** model to standardize network communication into seven layers:

1. **Physical:** Transmission of raw bit streams over a physical medium.
2. **Data Link:** MAC addressing and error detection (Frames).
3. **Network:** Routing and logical addressing (IP Addresses/Packets).
4. **Transport:** End-to-end communication and error recovery (TCP/UDP).
5. **Session:** Managing and terminating sessions between applications.
6. **Presentation:** Data encryption, compression, and translation.
7. **Application:** Network services for end-user applications (HTTP, FTP, SMTP).

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5. Network Devices

- **Network Interface Card (NIC):** Provides a physical connection to the network and holds the **MAC Address**.
- **Hub:** A "dumb" device that broadcasts data to all ports.
- **Switch:** An intelligent device that sends data only to the specific destination MAC address.
- **Router:** Connects different networks (e.g., your home LAN to the Internet) using IP addresses.
- **Modem:** Modulates/Demodulates signals between digital and analog formats.
- **Gateway:** A node that connects two networks using different protocols.

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6. IP Addressing (IPv4 vs. IPv6)

- **IPv4:** 32-bit address represented in four decimal parts (e.g., `192.168.1.1`). It allows for ≈4.3 billion addresses.

- **IPv6:** 128-bit address represented in hexadecimal (e.g., `2001:0db8:85a3...`). Created to solve the exhaustion of IPv4 addresses.

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7. MAC vs. IP Addresses

- **MAC Address (Physical):** Burned into the NIC; 48-bit; used for communication within the *same* local network.
- **IP Address (Logical):** Assigned by software/DHCP; used for communication *between* different networks.